[DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING]
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**A MAJOR PROJECT REPORT ON**

# RESEARCHMATE AI: AN AUTONOMOUS AI-POWERED WEB RESEARCH ASSISTANT


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## ABSTRACT

This project report details **ResearchMate AI**, an autonomous multi-agent research assistant designed to conduct grounded scientific literature synthesis from a single user topic. Built on Python, Streamlit, Google Gemini API, and Tavily Web Search API, the system automates topic decomposition into four objectives and formal research questions (RQ1–RQ4), multi-angle public web searching, adaptive query deepening, HTML extraction via BeautifulSoup, source authority assessment, grounded empirical claim extraction, literature review synthesis with in-text citations, 7-dimension research gap discovery, structured methodology design, and publication-grade export (17-section PDF report and 12-slide PowerPoint presentation). Empirical testing across three scientific domains confirms that the system maintains 100% bibliographic URL verifiability, eliminates citation hallucination, and delivers complete dossiers within 15–25 seconds.

## 1. Introduction

In modern scientific inquiry, the exponential growth of technical literature presents a formidable cognitive bottleneck. Conversational AI chatbots hallucinate citations and lack live web scraping tools. ResearchMate AI bridges this gap through a verified autonomous agent pipeline.

## 2. Problem Statement

Manual literature reviewing takes weeks, while standard LLMs hallucinate non-existent authors, papers, and DOIs. A verifiable, tool-using research assistant is required to automate evidence mapping.

## 3. Project Objectives

To implement an autonomous multi-agent pipeline; to query Tavily API with adaptive deepening; to anchor claims to direct quotes; to assess source credibility; to discover 7-dimension research gaps; and to compile publication-grade PDF and PPTX deliverables.

## 4. System Architecture & Workflow

The system coordinates specialized modules: Planner, ToolRegistry, Web Searcher, BeautifulSoup Extractor, Source Assessor, Evidence Engine, Literature Reviewer, Gap Finder, Methodology Designer, Citation Validator, and Deliverables Engine.

## 5. Technology Stack

Python 3.13, Streamlit 1.35+, Google Gemini API (google-genai), Tavily Search API, BeautifulSoup4, ReportLab, python-pptx, and SQLite 3.

## 6. Web Research & Extraction Methodology

Tavily search executes 3 to 6 targeted queries across diverse aspects. When sources are under 4, secondary queries deepen coverage. BeautifulSoup cleans paragraphs, stripping scripts and clutter.

## 7. Source Quality Assessor & Evidence Engine

Sources are classified into credibility tiers (ArXiv, IEEE, NIST, Web Blogs) with authority scores (0.0 to 1.0). The Evidence Engine anchors empirical claims to exact extracted passages with confidence scores.

## 8. 7-Dimension Research Gap Discovery

Each gap details: Gap ID, Title, Structural Explanation, Supporting Sources, Operational Impact, Suggested Research Question, Possible Methodology, and Feasibility.

## 9. Proposed Methodology & Validation Suite

The system designs a formal research framework with hypotheses (H1–H3), dataset requirements, variables, and proposed validation experiments clearly distinguished as proposals pending empirical execution.

## 10. Testing Results & Verification

All 8 automated tests passed in pytest. End-to-end multi-topic tests across Healthcare Cybersecurity, LLM Quantization, and Multi-Agent Disaster Response verified 100% URL verifiability and PDF/PPTX compilation.

## 11. Conclusion & Future Scope

ResearchMate AI proves that tool-using autonomous agents eliminate citation hallucination and automate literature synthesis. Future work includes direct PDF uploads and containerized sandbox experiment execution.